FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

FULL TEST – IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (07 – 09) contains 3 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (10 – 14) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.
3. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

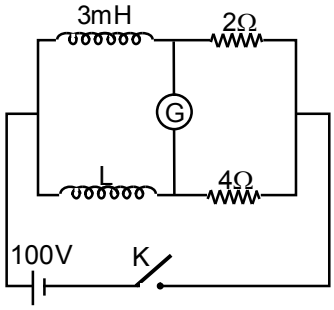
SECTION – A

Straight Objective Type

This section contains **6 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. A capacitor of capacitance C is connected between every two vertices of a cube (including the body diagonals and the face diagonals). The effective capacitance of the system across any edge is:
(A) C (B) 3C
(C) 2C (D) 4C

2. In the shown circuit, the switch is closed at t = 0. What should be the value of L if the galvanometer shows zero deflection?
(A) 3 mH
(B) 4 mH
(C) 6 mH
(D) 1.5 mH

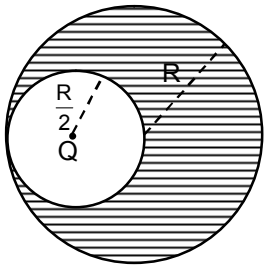


3. Two balls moving in a gravity free space collide with each other inelastically then choose the correct statement: (0 < e < 1)
(A) An any inertial frame the total pre collision kinetic energy is same.
(B) An any inertial frame the total post collision kinetic energy is same.
(C) The total velocity of separation of the balls will be different inertial frames.
(D) The loss in kinetic energy during collision is same in every inertial frame.

4. If I is the intensity at the central maximum of YDSE then what is the intensity at $y = \frac{\beta}{4}$?
(β is fringe width)
(A) $\frac{I}{4}$ (B) $\frac{I}{2}$
(C) 0 (D) $\frac{I}{3}$

Space for Rough work

5. An uncharged conducting sphere of radius R, has a hollow cavity sphere of radius $\frac{R}{2}$, as shown. Which of the following is/are correct about electrostatic energy?



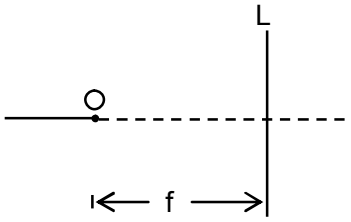
(A) Interaction energy of induced charges with charge placed at centre of cavity is $-\left(\frac{1}{4\pi\varepsilon_0}\right)\left(\frac{Q^2}{R}\right)$

(B) Interaction energy of induced charges with charge placed at centre of cavity is $\frac{1}{4\pi\varepsilon_0}\left(\frac{Q^2}{R}\right)$

(C) Electrostatic energy outside sphere is $\frac{1}{4\pi\varepsilon_0}\left(\frac{Q^2}{9R}\right)$

(D) Electrostatic energy outside sphere is $\left(\frac{1}{4\pi\varepsilon_0}\right)\left(\frac{Q^2}{R}\right)$.

6. A self luminous point object is placed at focus of a thin lens. Which of the following is INCORRECT about shape of wave fronts?
(A) Before lens it is spherical if lens is converging.
(B) Before lens it is spherical if lens is diverging.
(C) After lens it is spherical if lens is converging.
(D) After lens it is spherical if lens is diverging.

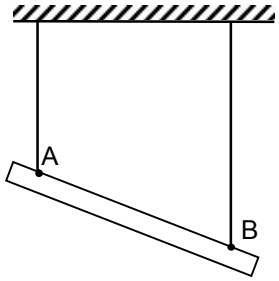


Space for Rough work

Multi Correct Choice Type

This section contains **3 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

7. Two vertical strings of different lengths are joined to a uniform rod at points A and B, rod is at rest. Now the string joining B is burnt. Which of the following are correct just after this?
(A) acceleration of COM of rod is vertical.
(B) acceleration of COM of rod is along rod.
(C) acceleration of particle at A is zero.
(D) acceleration of particle at A is horizontal.



8. A charged particle is projected in combined electric and magnetic field. Electric field is uniform along x-axis and magnetic field which is non-uniform but not time varying also along x-axis. Initial velocity of particle is $\left(10\hat{i}+10\hat{j}\right)$ m/s. After certain time x component of velocity is 20 m/s. Which of the following is correct at that time?
(A) Velocity of particle is $\left(20\hat{i}+10\hat{j}\right)$ m/s.
(B) Speed of particle is $10\sqrt{5}$ m/s.
(C) With given information we cannot calculate y component and z component of velocities.
(D) $V_y^2+V_z^2=100\,m^2/s^2$.

9. In a photo electric effect frequency of incident radiation is halved. As a result, what is possible?
(A) Kinetic energy of most energetic electrons, becomes less than half, if electrons are emitted.
(B) Kinetic energy of most energetic electrons becomes more than half, if electrons are emitted.
(C) Kinetic energy of most energetic electrons becomes half, if electrons are emitted.
(D) May be no photo electrons are obtained.

Space for Rough work

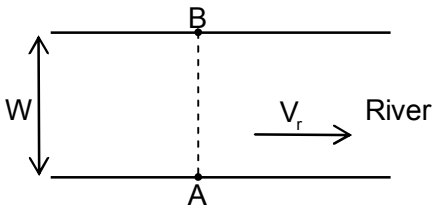
Comprehension type (Only One Option Correct)

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Question Nos. 10 and 11

A boatman starts from point A and always heads boat towards point B (i.e. V_{br} is always towards B).

V_{br} is the velocity of the boat in still water and W is the width of the river. It is given that $V_{br} > V_r$.



10. What is the time taken by the boat to cross the river?

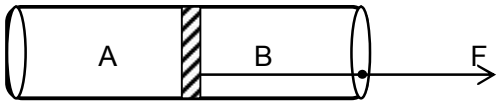
(A) $\frac{WV_r}{V_{br}^2}$ (B) $\frac{W}{V_{br} - V_r}$

(C) $\frac{WV_r}{V_{br}^2 - v_r^2}$ (D) $\frac{W}{\sqrt{V_{br}^2 - V_r^2}}$

11. If the boat is observed from the ground frame as it tries to reach point B:
(A) Its speed appears to increase with time
(B) Its speed appears to be constant with time
(C) Its speed appears to decrease with time
(D) None of the above.

Space for Rough work

Paragraph for Question Nos. 12 and 14



A ***diathermic*** piston free to move without friction divides an adiabatic cylinder in two equal parts. The two parts contain equal amount of ideal gas. Piston is joined to a string which comes out passing through right boundary. [Assume that no matter and energy flows through hole that passes the string]. Now the string is pulled slowly to right.

12. Temperature of gas A
(A) increases (B) Decrease
(C) first decrease then increase (D) first increase then decrease

13. Temperature of gas B
(A) increases (B) Decrease
(C) first decrease then increase (D) first increase then decrease

14. Work done by gas A is
(A) negative (B) positive
(C) zero (D) none of the above

Space for Rough work

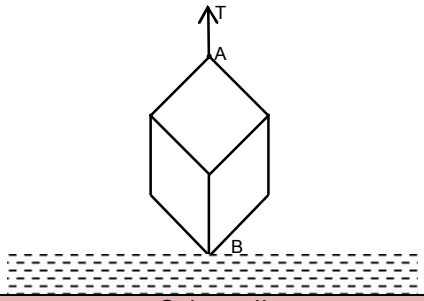
SECTION - B
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

	p	q	r	s	t
A	●	○	○	●	●
B	●	●	●	○	○
C	●	●	○	○	○
D	●	○	○	●	●

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. A cube of volume 1 m^3 and specific gravity 2 is very slowly being lowered in to a lake with help of a massless string tied to one of its vertices as shown.
Initially the vertex B is just touching the surface of water and finally, vertex A is just beneath the surface of water. For the process of immersion, match the columns: (Ignore change in KE of water). (g = 10 ms^{-2}).



Column I		Column II	
(A)	Work done by gravity on water + cube system.	(p)	$\frac{3\sqrt{3}}{2}\times10^4$ Joule.
(B)	Work done by tension on the cube.	(q)	$\frac{-3\sqrt{3}}{2}\times10^4$ Joule.
(C)	Work done by gravity on water.	(r)	$\frac{-\sqrt{3}}{2}\times10^4$ Joule.
(D)	Work done by gravity on the cube.	(s)	$2\sqrt{3}\times10^4$ Joule.
		(t)	$40\sqrt{3}\times10^4$ Joule.

2. Match the columns according to the particle released in the reaction:

Column I		Column II	
(A)	${}^{14}_{6}C \rightarrow {}^{14}_{7}N + e^- +$ ____	(p)	Neutrino
(B)	${}^{23}_{12}Mg \rightarrow {}^{23}_{11}Na + e^+ +$ ____	(q)	Anti Neutrino
(C)	${}^{81}_{36}Kr + e^- \rightarrow {}^{81}_{35}Br +$ ____	(r)	Deutron
(D)	${}^{137}_{55}Cs \rightarrow {}^{137}_{56}Ba + e^- +$ ____	(s)	α-particle.
		(t)	β-particle

Space for Rough work

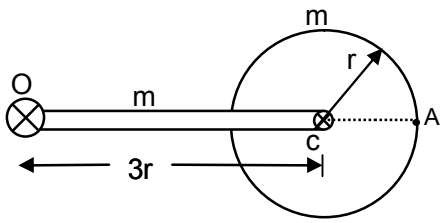
SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

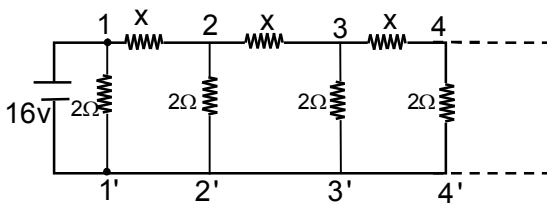
1. In a different type of Vernier calliper without zero error, five vernier division is equal to four main scale division. One main scale division is equal to 2 mm
Readings for a rod are.
Main scale reading = 8
Vernier scale reading = 3
If length of the rod is = [14 + x × 0.4] mm, what is the value of x?

2. A uniform rod of mass m and length 3r is free to rotate about a horizontal axis through O. A disc of mass m, radius r, is free to rotate about another horizontal axis fixed to other end of the rod (say C). [r = 1.2 m]. The system is released from rest when rod is horizontal. What is the velocity of particle at A (in m/s) when rod becomes vertical?

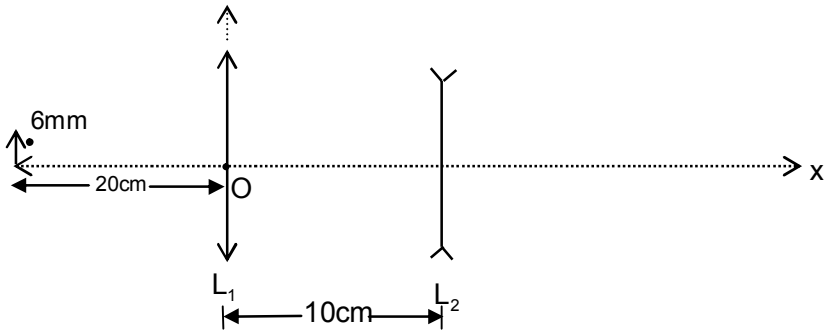


Space for Rough work

3. There is infinite ladder of network as shown. 1 and 1' are joined to ideal battery of emf 16 V. The potential difference across 4 and 4' is found to be 2V. What is the equivalent resistance between 1 and 1' (Voltage across the successive 2Ω resistance decreases by a factor of 2 as we move towards right)



4. A convex lens L_1 and a concave lens L_2 are arranged with separation 10 cm as shown. Their focal lengths are respectively 20 cm and 10 cm. optical centre of L_1 is treated as origin and optical axis as x-axis as shown. A self luminous point object is placed at [–20 cm, (6 mm)] Y coordinate of the final image in mm is?



5. Position vector of a particle is given as
$\vec{r} = A\cos^2 \omega t\hat{i} + B\sin^2 \omega t\hat{j}$, [A = 6 and B = 6]
Find the amplitude of motion.

Space for Rough work

Chemistry

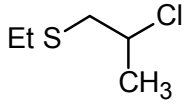
PART – II

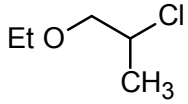
SECTION – A

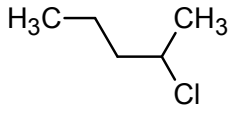
Straight Objective Type

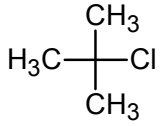
This section contains **6 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. Which will show greater degree of retention on hydrolysis?

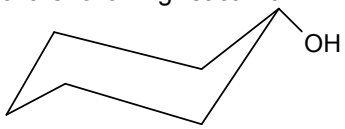
(A) 

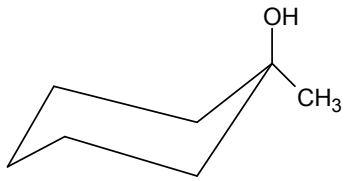
(B) 

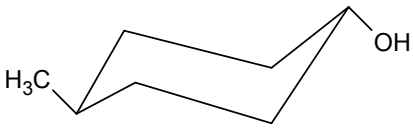
(C) 

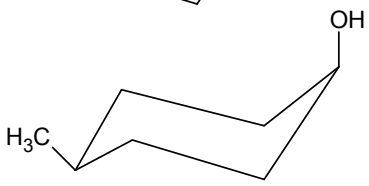
(D) 

2. Which of the following react with HBr at faster rate?

(A) 

(B) 

(C) 

(D) 

3. The value of universal gas constant R depends on
(A) Temperature of gas
(B) Volume of gas
(C) Number of moles of gas
(D) Units volume and pressure

Space for Rough work

4. When iodine is dissolved in aqueous potassium iodide, the shape of the species formed is:
(A) Linear (B) Angular
(C) Triangular (D) See-saw

5. Among the following complexes
(i) $[Ru(bipyridyl)_3)^+$
(ii) $[Cr(EDTA)]^+$
(iii) trans-$[CrCl_2(Oxalate)_2]^{3-}$
(iv) cis-$[CrCl_2(ox)_2]^{3-}$
The ones that shows chirality are
(A) (i),(ii), (iv) (B) (i), (ii), (iii)
(C) (ii), (iii), (iv) (D) (i), (iii), (iv)

6. The difference n^{th} and $(n + 1)^{th}$ Bohr's radius of H-atoms is equal to $(n-1)^{th}$ Bohr's radius. Hence the value of 'n' is:
(A) 1 (B) 2
(C) 3 (D) 4

Space for Rough work

Multi Correct Choice Type

This section contains **3 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

7. A binary solution of liquid A and B will show positive deviation from Roult's law if it fulfils the following condition:
   (A) $P_A > X_A P_A^o$ and $P_B > X_B P_B^o$
   (B) The intermolecular forces of A–B < A–A, B–B
   (C) ΔH mixing is positive
   (D) Δ V mixing is negative

8. For the reaction: $Cl_2(g) + 3F_2(g) \rightleftharpoons 2ClF_3(g)$; $\Delta H = -329KJ$ dissociation of $ClF_3(g)$ will be favoured by:
   (A) Increasing the temperature
   (B) Increasing the volume of the container
   (C) Adding of F_2 gas
   (D) Adding of inert gas at constant pressure

9. Select the correct statement(s):
   (A) When $T \to \infty$ or $E_a \to o$ then K = A
   (B) A positive catalyst can change ΔH of the reaction
   (C) A mixture of reactants may be thermodynamically unstable but kinetically stable
   (D) A negative catalyst increase the activation energy of the reaction

Space for Rough work

Comprehension type (Only One Option Correct)

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Question Nos. 10 and 11

Read the following paragraph and answer the questions given below:

A proper control by pH is very essential for many industrial as well as biological processes. Solutions with a definite pH can be prepared from single salts or mixtures of acids/bases and their salts. We also require solutions which resist change in pH and hence have a reserve value. Such solutions are called buffer solutions. Henderson gave a theoretical equal for preparing acidic buffers of definite pH. The equation is:

$$pH = pK_a + \log\frac{[Salt]}{[Acid]}$$

A similar equation is used for basic buffers. The pH of aqueous solution of single salts is calculated by using an expression whose exact form depends on the nature of the salt. Considering all that in mind, answer the following questions:

10. Calculate pH of a mixture containing 100 ml of 0.10 M NaOH mixed with 100 ml of 0.05 M CH_3COOH solution
(A) 10.4 (B) 11.7
(C) 12.4 (D) 13.4

11. Calculate pH if 100 ml of 0.1 M NaOH is mixed with 100 ml of 0.1 M CH_3COOH solution [K_a (CH_3COOH) = 1.8×10^{-5}) log 1.8 = 0.255]
(A) 8.71 (B) 7
(C) 6.25 (D) 9.37

Space for Rough work

Paragraph for Question Nos. 12 and 14

Molar conductivity ($\wedge_m$) is defined as conducting power of the ions produced by 1 mole of an electrolyte in a solution. $\wedge_m$ = k/c where κ is conductivity (in S cm^2 mol^{-1}) and c is molar concentration (in mole/cm^3) the molar conductivity of 0.04 M solution of $CaCl_2$ is 200 S cm^2 mol^{-1} at 298 K. A cell with electrode that are 2.0 cm^2 in surface area and 0.50 cm apart is filled with $CaCl_2$ solution.

12. Conductance of $CaCl_2$ solution is:
(A) 8×10^{-3} S (B) 32 S
(C) 0.032 S (D) None of these

13. How much current will flow when the potential difference between the two electrode is 5.0 V?
(A) 156.25 A (B) 0.16 A
(C) 160 A (D) None of these

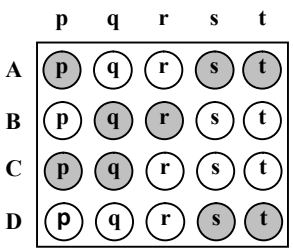
14. The cell constant of the following cell will be
(A) 0.4 (B) 0.8
(C) 0.25 (D) 0.6

Space for Rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the contents of Column – I with contents of Column – II for 1 mole of gas.

Column – I		Column – II	
(A)	Raw egg $\longrightarrow$ Boiled egg	p.	ΔH = + Ve
(B)	$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$	q.	ΔS = +Ve
(C)	$P_{(white)\ solid} \longrightarrow P_{(red)\ solid}$	r.	ΔG = –Ve
(D)	$Pd + H_2 \rightleftharpoons$ adsorbed gas	s.	Allotropic change
		t.	ΔS = – Ve

2. Match the contents of Column – I with contents of Column – II for 1 mole of gas.

Column – I		Column – II	
(A)	Tetragonal	(p)	$a \neq b \neq c$
(B)	Monoclinic	(q)	$\alpha \neq \beta \neq \gamma \neq 90^\circ$
(C)	Rhombohedral	(r)	$a \neq b \neq c$, $\alpha \neq \beta \neq \gamma \neq 90^\circ$
(D)	Triclinic	(s)	$a = b \neq c$, $\alpha \neq \beta \neq \gamma \neq 90^\circ$
		(t)	$a = b = c$, $\alpha \neq \beta \neq \gamma \neq 90^\circ$

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. An α-particle moving with velocity $\frac{1}{30}$ th times of velocity of light. If uncertainty in position is $\frac{3.31}{\pi}$ pm, then minimum Uncertainty in kinetic energy is y × 10^{-16}J. Calculate value of 'y'.

2. The vander Waal's constants for a gas are a = 3.6 atm L^2 mol^{-2}, b = 0.6L mol^{-1}, If R = 0.08 L $atmk^{-1}$ mol^{-1}. If the Boyle's temperature (K) is T_b of this gas, then what is the value of $\frac{T_b}{15}$?

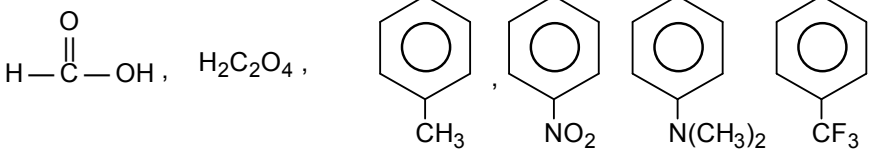
3. Amongst the following, the total number of compounds whose aqueous solution turns red litmus paper blue is NaCN, KCl, CH_3COONH_4, NaH_2PO_4, $ZnCl_2$, Na_3PO_4, $Fe(NO_3)_3$, Na_2CO_3, NH_4Cl, $NaHCO_3$, $Na_2C_2O_4$, Na_2HPO_4 .

Given :

Acid	Ka_1	Ka_2	Ka_3
H_3PO_4	10^{-3}	10^{-8}	10^{-12}
H_2CO_3	10^{-6}	10^{-1}	-
$H_2C_2O_4$	10^{-2}	10^{-5}	-

4. $\Delta G°$ for the hypothetical reaction x + y $\rightleftharpoons$ z is – 4.606 kcal. The equilibrium constant of the reaction at 227° is (x × 10^2). Find the value of x.

5. Find out number of compounds which are more acidic than benzoic acid from the following



Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **6 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. A straight line 'L' cuts the lines AB, AC and AD of a parallelogram ABCD at points B_1, C_1 and D_1 respectively. If $\overrightarrow{AB_1} = \lambda_1\overrightarrow{AB}$, $\overrightarrow{AD_1} = \lambda_2\overrightarrow{AD}$ and $\overrightarrow{AC_1} = \lambda_3\overrightarrow{AC}$, then $\frac{1}{\lambda_3}$ is equal to

(A) $\frac{1}{\lambda_1}+\frac{1}{\lambda_2}$ (B) $\frac{1}{\lambda_1}-\frac{1}{\lambda_2}$

(C) $-\lambda_1 + \lambda_2$ (D) $\lambda_1 + \lambda_2$

2. If $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then det $(f(\theta + \phi))$ is

(A) 0 (B) 1

(C) –1 (D) $\cos^2\theta - \sin^2\phi$

3. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non zero and non coplanar vectors and $\vec{p}, \vec{q}$ and $\vec{r}$ be three vectors given by $\vec{p} = \vec{a} + \vec{b} - 2\vec{c}$; $\vec{q} = 3\vec{a} - 2\vec{b} + \vec{c}$ and $\vec{r} = \vec{a} - 4\vec{b} + 2\vec{c}$. If the volume of the parallelepiped determined by $\vec{a}, \vec{b}$ and $\vec{c}$ is v_1 and that of the parallelepiped determined by $\vec{p}, \vec{q}$ and $\vec{r}$ is v_2 then $v_2 : v_1$ is

(A) 1 : 5 (B) 5 : 1

(C) 15 : 1 (D) 1 : 15

4. The point A divides the join of P(–5, 1) and Q(3, 5) in ratio K : 1. The two values of K for which area of ΔABC, where B is (1, 5) and C is (7, –2), is equals to 2 units in magnitude are K or $\frac{31}{K+2}$ then K is

(A) 2 (B) 3

(C) 5 (D) 7

Space for rough work

5. If $\frac{\sin x}{a}=\frac{\cos x}{b}=\frac{\tan x}{c}=k$, then $bc+\frac{1}{ck}+\frac{ak}{1+bk}$ is equal to

(A) $k\left(a+\frac{1}{a}\right)$ (B) $\frac{1}{k}\left(a+\frac{1}{a}\right)$

(C) $\frac{1}{k^2}$ (D) $\frac{a}{k}$

6. The area (Δ) and an angle (θ) of a triangle are given. When the side opposite to the given angle is minimum, then the lengths of the remaining two sides are

(A) $\sqrt{\frac{2\Delta}{\sin\theta}}, \sqrt{\frac{3\Delta}{\sin\theta}}$ (B) $\sqrt{\frac{2\Delta}{\sin\theta}}, \sqrt{\frac{2\Delta}{\sin\theta}}$

(C) $\sqrt{\frac{4\Delta}{\sin\theta}}, \sqrt{\frac{2\Delta}{\sin\theta}}$ (D) $\sqrt{\frac{6\Delta}{\sin\theta}}, \sqrt{\frac{6\Delta}{\sin\theta}}$

Multi Correct Choice Type

This section contains **3 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

7. Given a real valued function f such that $f(x)=\begin{cases}\frac{\tan^2\{x\}}{x^2-[x]^2} & \text{for } x>0\\ 1 & \text{for } x=0\\ \sqrt{\{x\}\cot\{x\}} & \text{for } x<0\end{cases}$ (where [.] denotes the greatest integer function and {x} is fractional part of x), then

(A) $\lim\limits_{x\to0^+} f(x)=1$ (B) $\lim\limits_{x\to0^-} f(x)=\cot 1$

(C) $\cot^{-1}\left(\lim\limits_{x\to0^-} f(x)\right)^2=1$ (D) $\tan^{-1}\left(\lim\limits_{x\to0^+} f(x)\right)=\frac{\pi}{4}$

8. The equation $x^{\left[(\log_3 x)^2-\left(\frac{9}{2}\right)\log_3 x+5\right]}=3\sqrt{3}$ has

(A) at least 1 real solution (B) exactly three real solution

(C) exactly 1 irrational solution (D) complex roots

9. Which of following functions have the maximum value unity?

(A) $\sin^2 x-\cos^2 x$ (B) $\sqrt{\frac{6}{5}}\left(\frac{1}{\sqrt{2}}\sin x+\frac{1}{\sqrt{3}}\cos x\right)$

(C) $\cos^6 x+\sin^6 x$ (D) $\cos^2 x+\sin^4 x$

Space for rough work

Comprehension type (Only One Option Correct)

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Question Nos. 10 and 11

Read the following write up carefully and answer the following questions:

Let f(x) be a non-positive continuous function and $F(x)=\int_0^x f(t)dt \ \forall\ x \geq 0$ and $f(x) \geq cF(x)$ where c > 0 and let g: $[0, \infty) \to R$ be a function such that $\frac{dg(x)}{dx} < g(x) \ \forall\ x > 0$ and g(0) = 0

10. The total number of root(s) of the equation f(x) = g(x) is/are
(A) ∞ (B) 1
(C) 2 (D) 0

11. The number of solution(s) of the equation $|x^2 + x - 6| = f(x) + g(x)$ is/are
(A) 2 (B) 1
(C) 0 (D) 3

Paragraph for Question Nos. 12 and 14

Read the following write up carefully and answer the following questions:
If 4-letter words are formed using the letters of the word 'MORADABAD', then

12. The probability that at least two letters are repeated is
(A) $\frac{120}{313}$ (B) $\frac{133}{313}$
(C) $\frac{123}{310}$ (D) none of these

13. The probability that D comes in the 4-letter word is
(A) $\frac{209}{313}$ (B) $\frac{120}{313}$
(C) $\frac{189}{310}$ (D) none of these

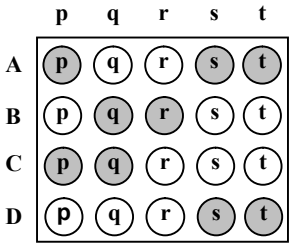
14. The probability that D comes exactly once in the 4-letters word is
(A) $\frac{73}{313}$ (B) $\frac{144}{310}$
(C) $\frac{146}{313}$ (D) none of these

Space for rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	The number of solutions of the equation $\lvert\cot x\rvert = \cot x + \frac{1}{\sin x}(0 < x < \pi)$ is	(p)	No solution
(B)	If $\sin\theta + \sin\phi = \frac{1}{2}$ and $\cos\theta + \cos\phi = 2$, then value of $\cot\left(\frac{\theta+\phi}{2}\right)$ is	(q)	$\frac{1}{3}$
(C)	The value of $\sin^2\alpha + \sin\left(\frac{\pi}{3}-\alpha\right)\sin\left(\frac{\pi}{3}+\alpha\right)$ is	(r)	1
(D)	If $\tan\theta = 3\tan\phi$, then maximum value of $\tan^2(\theta-\phi)$ is	(s)	2
		(t)	$\frac{3}{4}$

2. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	$\int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}}dx$	(p)	$\ln\left(\frac{(x^2+1)+\sqrt{x^4+1}}{x}\right)+C$
(B)	$\int \frac{x^2-1}{x\sqrt{1+x^4}}dx$	(q)	$C+\frac{1}{\sqrt{2}}\ln\left(\frac{\sqrt{x^4+1}-\sqrt{2}x}{(x^2-1)}\right)$
(C)	$\int \frac{1+x^2}{(1-x^2)\sqrt{1+x^4}}dx$	(r)	$C-\tan^{-1}\left(\sqrt{\sqrt{1+\frac{1}{x^4}}-1}\right)$
(D)	$\int \frac{dx}{(1+x^4)\sqrt{\sqrt{1+x^4}-x^2}}$	(s)	$\frac{\sqrt{x^4+x^2+1}}{x}+C$
		(t)	$\ln\left(\frac{(x^2+1)-\sqrt{x^4+1}}{x}\right)+C$

Space for rough work

SECTION – C
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. If $x = \frac{4}{9}$ satisfies the in-equation $\log_a(x^2 - x + 2) > \log_a(-x^2 + 2x + 3)$, then sum of all possible distinct values of [x] is (where [.] denotes the greatest integer function) is _____

2. Let $\vec{v} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{w} = \hat{i} + 3\hat{k}$. If $\vec{u}$ is a unit vector and P is the maximum value of the scalar triple product $[\vec{u}\ \vec{v}\ \vec{w}]$, then the value of $p^2 - 50$ is _____

3. If $\int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}}dx = \mu e^x\left(\frac{1+x}{1-x}\right)^{\lambda} + c$, then $2(\lambda + \mu)$ is equal to _____

4. If A, B and C are the angles of a triangle and $\begin{vmatrix} \sin 2A & \sin C & \sin B \\ \sin C & \sin 2B & \sin A \\ \sin B & \sin A & \sin 2C \end{vmatrix} = \lambda \sin A \sin B \sin C$, then λ is _____

5. The number of solutions of the equation $\log_{|x-1|}|x^2 - 1| = [|\sin x| + |\cos x|]$ (where [.] denotes the greatest integer function) is _____

Space for rough work